Caption-Image Retrieval

Searching an image collection with natural language

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1. Motivation

- Enormous numbers of images are produced and stored every day
- Finding *the right* image by scrolling is painful
- What if we could search images the way we search text?

> *Type a sentence → get the images that match it.*

2. Problem Statement

Input

- A collection of raw images
- A natural-language query describing a desired image

Output

- A ranked list of the **top-K** most relevant images

Framing

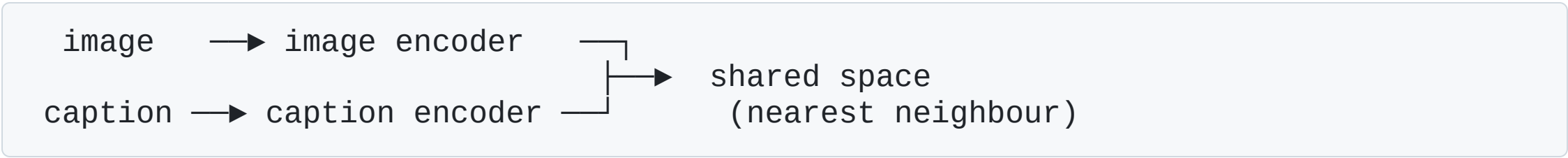
- Learn a **shared embedding space** where captions and images are aligned
- Retrieval = nearest-neighbour search in that space

3. Dataset — Microsoft COCO

- **123,287** images, **5** human-annotated captions each
- Karpathy & Fei-Fei (2015) split:
  - 113,287 train · 5,000 val · 5,000 test
- **Images:** precomputed 10-crop VGG-19 `fc7` features → 4,096-d per image
- **Text:** GloVe 300-d word vectors

We train on a 5,000-image subset of the training split (CPU budget).

4. Approach — a shared space



- Similar pairs → **close** in the space
- Similarity = **negative squared Euclidean distance**
- Retrieval: embed the query, return the nearest image embeddings

5. Baseline — multi-response linear regression

- Caption = **mean of its GloVe word vectors**
- Fit a weight matrix `W` so that `X @ W ≈ Y`
  - `X` = caption features, `Y` = VGG-19 image features
- Least squares — **closed form**, no gradient descent

Weakness: mean-pooling **discards word order** and dilutes the object being described.

6. Model — GRU + GloVe

- **Image encoder:** learned linear map `f_i(i) = W_i · f_VGG(i)`
- **Caption encoder:** GRU over GloVe-initialized embeddings, `f_c(c) = GRU(c)`
- Embeddings **L2-normalized**
- **Margin ranking loss:**
  - correct pair must score better than wrong pairs by a margin
  - negatives come free from the same batch

Following Vendrov et al. (2015), "Order-Embeddings of Images and Language".

7. Implementation

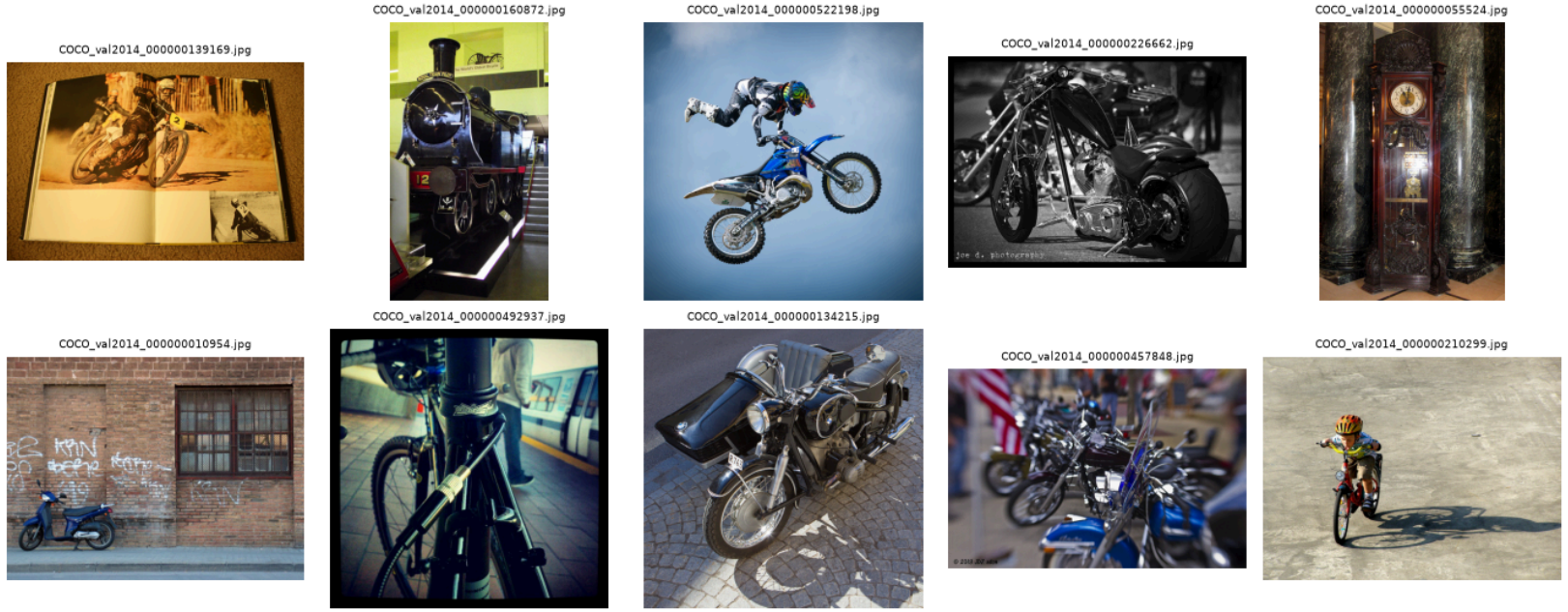
- **NumPy** — features, GloVe, baseline
- **PyTorch** — GRU + image projection, trained on **CPU**
- `src/` modules:
  - `data.py` `glove.py` `baseline.py` `run_baseline.py`
  - `vocab.py` `models.py` `train_rnn.py`
  - `retrieve.py` `demo.py`
- Trained on **AWS EC2** `t3.xlarge`, 30 epochs, ~41 s/epoch

8. Results

Model	R@1	R@5	R@10	Mean rank
Baseline	2.8	9.9	16.7	179.4
GRU + GloVe	**5.0**	**18.0**	**28.3**	**85.6**
Vendrov et al. (reported)	37.0	—	86.8	7.3

- GRU beats the baseline on **every metric**
- R@10: **16.7% → 28.3%** · mean rank: **179 → 86**
- Absolute gap vs reported: training scale (5k subset / 30 epochs vs 113k / 50)

9. Demo — query: "a cycle"



- 8 of the top 10 results are **bikes / motorcycles / scooters**
- The model has **never seen the word "cycle"** paired with these images

Top hit: "two magazine photos of a motorcycle and rider".

10. Training visibly improves retrieval

Same query "a cycle" — epoch ~20 vs epoch 30:

	epoch ~20	epoch 30
cycle-related in top 10	5	**8**
new correct results	—	motor scooter, parked motorcycles, boy on bike

- The gain is not just the aggregate R@10 — the **rankings get cleaner**

11. Honest failures

- A few irrelevant results survive in the top 10:
  - *"a little locomotive in a museum"*
  - *"a large clock between two pillars"*
- **Why:**
  - trained on a 5,000-image subset
  - image side is a single **global** VGG vector — no attention over regions
  - captions are short; fine-grained composition is hard

Showing the misses is more useful than pretending the model is perfect.

12. Conclusions & future work

Conclusions

- Reproduced the paper's core claim at smaller scale
- Baseline fails on multi-object captions; the **GRU fixes it**
- R@10 **28.3%**, mean rank **86** — both models still improving at epoch 30

Future work

- **Attention** over image regions
- Train on the **full 113k split**
- Compare against the **order-embedding** objective of Vendrov et al.

Thank you

Questions?

`github.com/Vishanth21/neural_caption-image_retrieval`